生物统计和生物信息引论

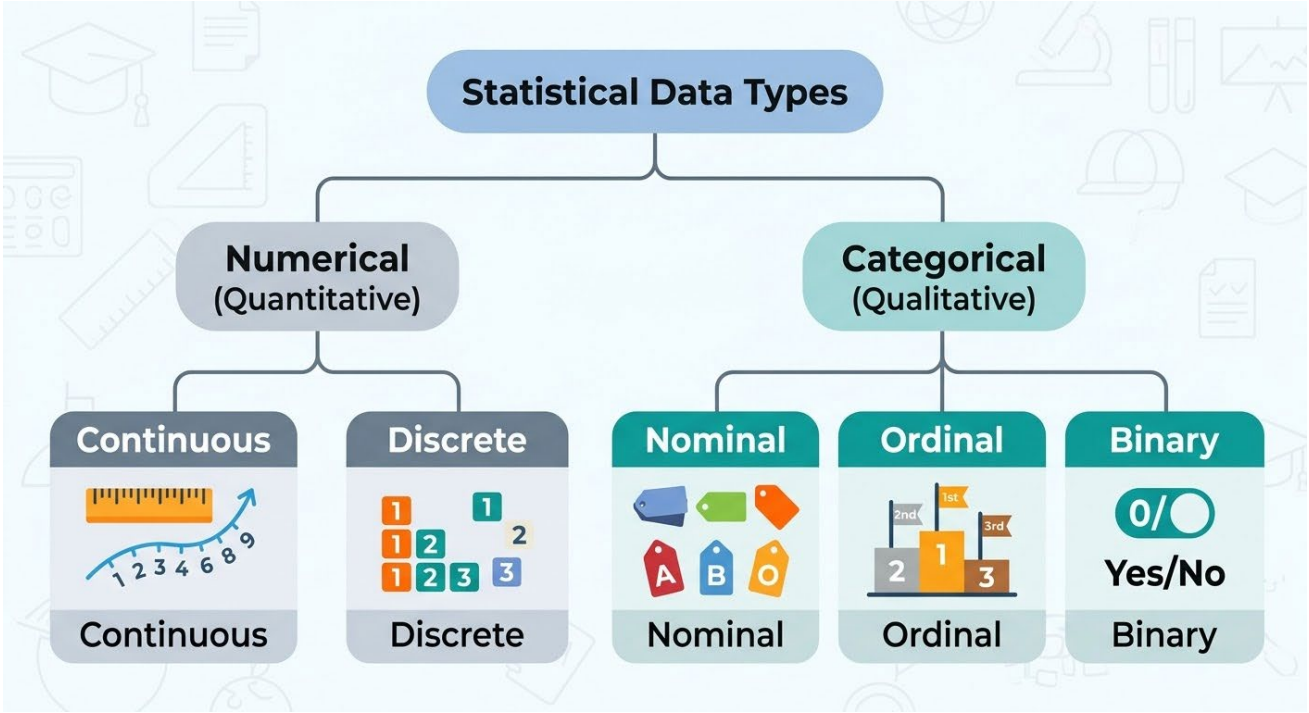
Lecture 5

类别变量与非参数方法（I）

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Recall：数据类型

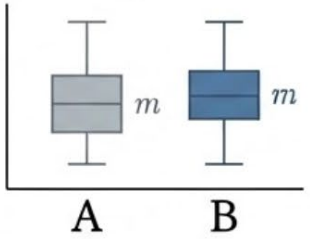
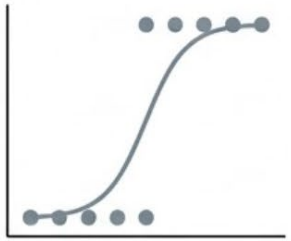
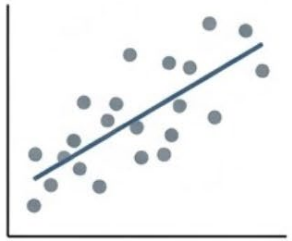
1. **Numerical** (数值型) - Quantitative
    - **Continuous** (连续型): Height, Weight, Gene Expression ($y \in \mathbb{R}$)
    - **Discrete** (离散型): Counts such as # cigarettes smoked, # car accidents ($y \in \mathbb{N}$)
2. **Categorical** (类别型) - Qualitative
    - **Binary** (二分类): Yes/No, Dead/Alive
    - **Nominal** (无序分类): Blood Type (A, B, AB, O)
    - **Ordinal** (有序分类): Cancer Stage (I, II, III, IV)



Predictor vs. Outcome

Statistical method is strictly dictated by the data types of your Predictor X and Outcome Y.

OUTCOME (Y)

PREDICTOR (X)	Categorical	Continuous
Categorical	Categorical X, Categorical Y • Contingency Table Analysis 	Categorical X, Continuous Y • ANOVA (F) 
Continuous	Continuous X, Categorical Y • Logistic Regression • Discriminant Analysis 	Continuous X, Continuous Y • Correlation (r) • Linear Regression (R^2) 

Notes: R^2 is the coefficient of determination; F denotes the test statistic.
This matrix provides a general methodological guide based on data types.

- Focus on the simplest setting
- X in {0, 1}, Y in {0, 1}
- Observed data

ID	X	Y
1	0	1
2	0	0
3	0	0
4	0	0
5	1	0
6	0	0
7	1	1
8	0	0
9	0	1
…	…	…

2 × 2列联表（Contingency Table ）

- Contingency Table = Classification Table: **frequency** of outcomes
- Two-way 2x2 table: frequency outcomes of two categorical variables

		Lung Cancer		
		Cases	Controls	Total
Smoking	Yes	688	650	1338
	No	21	59	80
	Total	709	209	1418

- Binary predictor (X)
  - ▶Smoking
- Binary outcome (Y)
  - ▶Lung Cancer
- Counts
  - ▶# Observations

Joint / Conditional / Marginal Distribution

Underlying probability distribution of X and Y

1. **Joint distribution**: Probability of a specific cell (i, j):

$$\pi_{ij} = P(X = i, Y = j)$$

2. **Marginal distribution**:

- Row Marginals (Probability of X):

$$P(X = 1) = \pi_{1+} = \pi_{11} + \pi_{12}$$

$$P(X = 2) = \pi_{2+} = \pi_{21} + \pi_{22}$$

- Column Marginals (Probability of Y):

$$P(Y = 1) = \pi_{+1} = \pi_{11} + \pi_{21}$$
$$P(Y = 2) = \pi_{+2} = \pi_{12} + \pi_{22}$$

	Column		
Row	1	2	Total
1	π_{11} $(\pi_{1\|1})$	π_{12} $(\pi_{2\|1})$	π_{1+} (1.0)
2	π_{21} $(\pi_{1\|2})$	π_{22} $(\pi_{2\|2})$	π_{2+} (1.0)
Total	π_{+1}	π_{+2}	1.0

$$\pi_{11} + \pi_{12} + \pi_{21} + \pi_{22} = 1$$

Homogeneity of conditional distributions = independence

P(Y=j | X=i) = P(Y = j)

3. **Conditional distribution**: Distribution of Y given X:

$$P(Y = i|X = j) = \pi_{i|j} = \frac{P(Y = i, X = j)}{P(X = j)} \qquad \pi_{1|1} = \frac{\pi_{11}}{\pi_{1+}}$$

How the 2×2 Table is Collected

	$Y = 1$	$Y = 2$	Row total
$X = 1$	n_{11}	n_{12}	n_{1+}
$X = 2$	n_{21}	n_{22}	n_{2+}
Col total	n_{+1}	n_{+2}	N

- **Fix X， observe Y**

  e.g. In cohort study, sample 500 with $X = 1$ (smoking) and 500 with $X = 2$(non-smoking)

  follow-up and compare the proportion of $Y = 1$ (lung cancers).

- **Fix Y， observe X**

  e.g. In case-control study, collect 500 with $Y = 1$ (rare disease) and 500 with $Y = 2$(healthy)

  follow-up and compare the proportion of $X = 1$ (exposure).

- **Hypergeometric sampling:** Both row and column margins are fixed

  e.g. Lady testing tea： 8 cups, tea (4) or milk (4) added first. Pick which 4.

Sampling scheme and sampling distribution

	$Y = 1$	$Y = 2$	Row total
$X = 1$	n_{11}	n_{12}	n_{1+}
$X = 2$	n_{21}	n_{22}	n_{2+}
Col total	n_{+1}	n_{+2}	N

- **Fix X， observe Y**

$$n_{11} \mid n_{1+} \sim \mathrm{Bin}(n_{1+}, \pi_{1|1}), \qquad n_{21} \mid n_{2+} \sim \mathrm{Bin}(n_{2+}, \pi_{1|2}).$$

- **Fix Y， observe X**

$$n_{11} \mid n_{+1} \sim \mathrm{Bin}\left(n_{+1}, \frac{\pi_{11}}{\pi_{+1}}\right), \qquad n_{12} \mid n_{+2} \sim \mathrm{Bin}\left(n_{+2}, \frac{\pi_{12}}{\pi_{+2}}\right).$$

- **Hypergeometric sampling:** Both row and column margins are fixed. One degree of freedom n_{11}

$$n_{11} \sim \mathrm{Hypergeometric}(N, n_{1+}, n_{+1}),$$

$$P(n_{11} = k) = \frac{\binom{n_{1+}}{k}\binom{n_{2+}}{n_{+1}-k}}{\binom{N}{n_{+1}}}.$$

Measuring Association (1): Difference of proportions

- **Difference of proportions (d)**: $P(Y=1|X=1)$ - $P(Y=1|X=2)$
- 暴露组与非暴露组发生事件的绝对概率差
- Point Estimate: $\hat{d} = n_{11}/n_{1+} - n_{21}/n_{2+}$
- Interpretation: The absolute change in risk attributed to the exposure.

  $d = 0$ implies no difference

  $d > 0$ means the exposure increases the risk

  $d < 0$ means it is protective
- The "Public Health" Perspective: Suppose a rare disease increases from 0.001% to 0.002% The risk doubled, but the absolute difference is only 0.001%.

Measuring Association (2): Relative Risk (RR)

If the incidence rate for two groups increases from 50% to 51% and from 1% to 2%, respectively.

The absolute difference is 49% in both cases. But are these two situations equally serious?

$P(Y = 1|X = 1)$ = 50%

$P(Y = 1|X = 2)$ = 1%

$P(Y = 1|X = 1)$ = 51%

$P(Y = 1|X = 2)$ = 2%

Measuring Association (2): Relative Risk (RR)

- **Relative risk(*RR*)**: $RR = \frac{P(Y=1|X=1)}{P(Y=1|X=2)}$

暴露组发病风险与非暴露组发病风险的**比值**

- Point Estimate: $\hat{RR} = \frac{\hat{P}(Y=1|X=1)}{\hat{P}(Y=1|X=2)} = \frac{n_{11}/n_{1.}}{n_{21}/n_{2.}}$
- $RR = 1$: no association
- It tells us how many times more likely an event is to occur due to the exposure.
- Invariance under $\frac{c\,P(Y=1|X=1)}{c\,P(Y=1|X=2)}$
- Comparing across populations / strata ; Focus is on strength of association

Measuring Association (3): Odds Ratio (OR)

- **Odds:** 发生概率 / 不发生概率 (p/(1 -p))
  - Odds In the group X= 1 $\dfrac{P(Y=1|X=1)}{1-P(Y=1|X=1)}$
- **Odds ratio(OR):** The ratio of two odds comparing two groups**.**

$$\theta = \frac{P(Y=1|X=1)}{1-P(Y=1|X=1)} \Big/ \frac{P(Y=1|X=2)}{1-P(Y=1|X=2)} = \frac{P(Y=1,X=1)P(Y=2,X=2)}{P(Y=1,X=2)P(X=1,Y=2)} = \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}}$$

- Point Estimate: $\hat{\theta} = \dfrac{n_{11}n_{22}}{n_{12}n_{21}}$
- $\log\theta$ is **symmetric ;** $\theta = 1$ means **no association,**
- Odds ratio **doesn't change** when table is transposed.

Measuring Association (3): Odds Ratio (OR)

- **Zero Cell Problem**: When some $n_{ij} = 0$, θ is not a good estimator.

  Can be improved by adding 0.5 to each cell $\tilde{\theta} = \dfrac{(n_{11}+0.5)(n_{22}+0.5)}{(n_{12}+0.5)(n_{21}+0.5)}$.

ORIGINAL 2 × 2 TABLE
(Zero Count)

	Yes	No	Totals
X_1	0	10	21
X_2	1	10	21
Totals	1	20	21

+0.5 to *each* cell count →

CORRECTED 2 × 2 TABLE
(+0.5 Adjustment)

	Yes	No	Totals
X_1	0.5	10.5	23
X_2	1.5	10.5	23
Totals	2.0	21.0	23

Recall：Case-Control Study（病例对照研究）

- 由果溯因：按照患病状态分组，回溯暴露情况
- 流程：选取已经患病的人作为“病例组”（Case），选取未患病但背景相似的人作为“对照组”（Control）。然后回头调查他们过去的暴露史。
- 直觉：肺癌患者中，吸烟的比例是不是比健康人更高。

- 适用于罕见病的回顾性研究

- Weaker evidence than cohort studies

- Case-Control Study: 按是否患肺癌选人，然后回顾是否吸烟

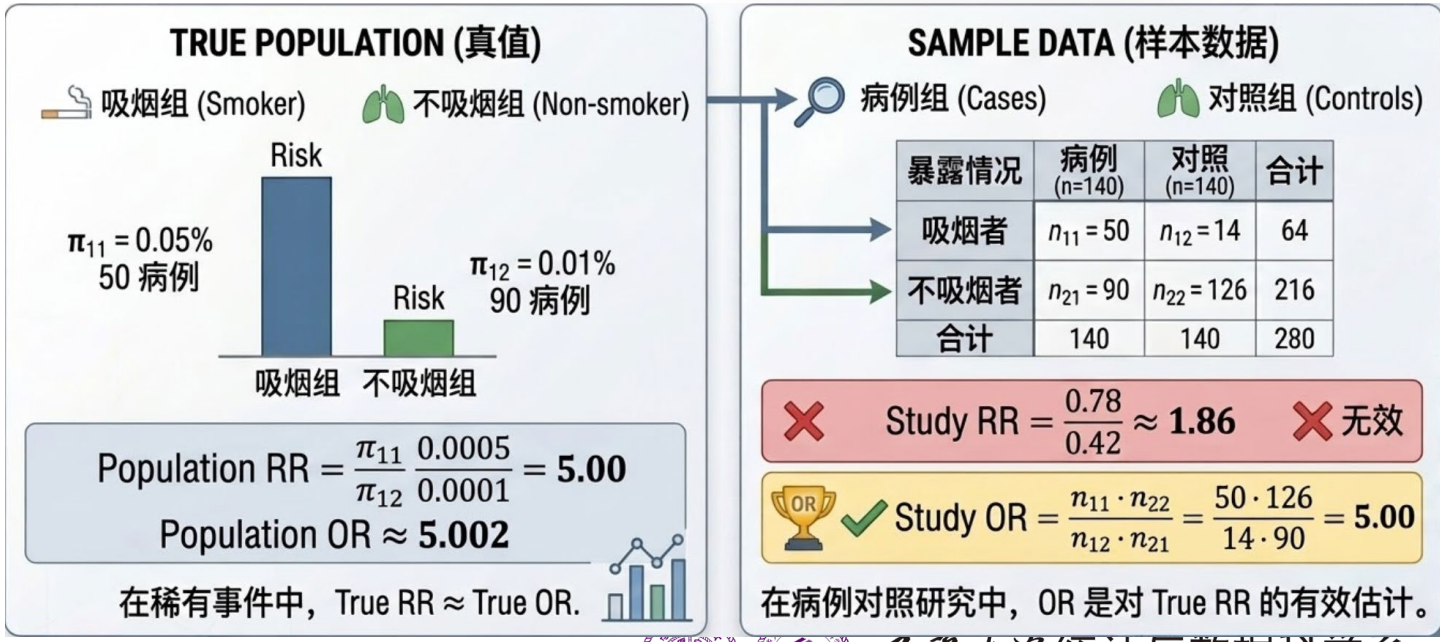
Case-Control Study

- Under case-control sampling, RR is not estimable (why?)
- Rare event assumption in both groups of X

Small $P(Y=1|X=1)$ and $P(Y=1|X=2)$ the odds ratio ≈ relative risk

$$OR = RR \times \frac{1-P(Y=1|X=2)}{1-P(Y=1|X=1)}$$

- Case-control OR = Population OR



Summary

- cohort/prospective design -> RD, RR are natural
- case-control design - > OR
- OR is invariable across case-control, follow-up or cross-sectional study
- Caution of OR for common events
- No specific number; for policy, we do not how many people benefit

Statistical inference

- Two-sample test for two binomial proportions
  - assume you are familiar with one sample binomial test
- Confidence intervals for the difference of proportions and relative risk

- 2 by 2 Contingency table
  - Chi-square test and Fisher's exact test

Two-Sample Test for Two Binomial Proportions

- Let X denote the row variable and Y denote the column variable in the 2x2 table.
- Conditional probabilities of interest:

$$p_1 = P(Y = 1 \mid X = 1), \qquad p_2 = P(Y = 1 \mid X = 2)$$

Two i.i.d. samples of sizes n_1 and n_2

$$Z_{1\ell} = I(Y_{1\ell} = 1) \sim \text{Bernoulli}(p_1), \qquad \ell = 1, \ldots, n_{1+},$$

$$Z_{2\ell} = I(Y_{2\ell} = 1) \sim \text{Bernoulli}(p_2), \qquad \ell = 1, \ldots, n_{2+}$$

Point estimator of the difference:

$$\hat{p}_1 - \hat{p}_2 = \frac{n_{11}}{n_{1+}} - \frac{n_{21}}{n_{2+}}$$

Our cell counts are discrete, but thanks to the Central Limit Theorem (CLT), when sample size n is large, our estimators approximate a Normal distribution.

$$\widehat{p}_1 - \widehat{p}_2 \overset{d}{\approx} \mathcal{N}\left(p_1 - p_2, \frac{p_1(1-p_1)}{n_{1+}} + \frac{p_2(1-p_2)}{n_{2+}}\right)$$

$$Var(\widehat{p}_1 - \widehat{p}_2) = \frac{p_1(1-p_1)}{n_{1+}} + \frac{p_2(1-p_2)}{n_{2+}}$$

This allows us to use the standard 95% CI formula: $z_{\left\{\frac{\alpha}{2}\right\}}$

$$\widehat{p}_1 - \widehat{p}_2 \pm 1.96 \left(\frac{\widehat{p}_1(1-\widehat{p}_1)}{n_{1+}} + \frac{\widehat{p}_2(1-\widehat{p}_2)}{n_{2+}}\right)^{1/2}$$

Hypothesis testing: $H_0: p_1 = p_2 = p$

NULL distribution $$\widehat{p}_1 - \widehat{p}_2 \overset{d}{\approx} \mathcal{N}\left(0, \frac{p(1-p)}{n_{1+}} + \frac{p(1-p)}{n_{2+}}\right)$$

Test statistics $$Z = (\widehat{p}_1 - \widehat{p}_2) / \left(\frac{\widehat{p}(1-\widehat{p})}{n_{1+}} + \frac{\widehat{p}(1-\widehat{p})}{n_{2+}}\right)^{1/2}$$

$$Z \sim N(0,1)$$

Remark: how to estimate p

Rejection region

Critical values $z_{\{\frac{\alpha}{2}\}}$ and $z_{1-\alpha/2}$

P−vlaue: $2 \times P(N(0,1) > |Z|)$

Example

- 2x2 contingency table (observed)

	Age at First Birth		
Breast Cancer Status	$\geq$30	<30	Total
With	683	2537	3220
Without	1498	8747	10245
Total	2181	11284	13465

- Whether the underlying probability of having an age at first birth of $\geq$30 is different in the two groups
  - $H_0 : p_1 = p_2$ versus $H_0 : p_1 \neq p_2$
  - p_1 and p_2 are the probabilities that age at first birth is $\geq$30 in women with and without breast cancer, respectively
  - H_0 : Age at birth is independent of breast cancer status

Example

- 683 of 3220 women with breast cancer and 1498 of 10245 women without breast cancer had an age at first birth $\geq$30
  - $n_1 = 3220$ and $n_2 = 10245$
  - $\widehat{p}_1 = \dfrac{683}{3220} = 0.212$ and $\widehat{p}_2 = \dfrac{1498}{10245} = 0.146$
  - $\widehat{p} = \dfrac{683 + 1498}{3220 + 10245} = 0.162$

$$z \approx 8.85, \qquad p\text{-value} \approx 8.54 \times 10^{-19}$$

Confidence intervals

- Ratios are severely skewed (bounded by 0 below, stretching to ∞ above). A normal CI would be inaccurate.
- We take the natural log, The log-transformed ratios $\ln(\theta)$ are beautifully symmetric around 0 and normally distributed.

2. Relative Risk (RR):

$$95\%\ \text{CI for}\ RR = \exp(\ln \hat{RR} \pm 1.96 \cdot SE(\ln \hat{RR}))$$

$$Var(\ln \hat{RR}) = \frac{1-p_1}{p_1 n_{1+}} + \frac{1-p_2}{p_2 n_{2+}}$$

3. Odds Ratio (OR):

$$95\%\ \text{CI for}\ OR = \exp(\ln \hat{\theta} \pm 1.96 \cdot SE(\ln \hat{\theta}))$$

$$SE(\ln \hat{\theta}) = \sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}}$$

Pearson 卡方检验 (Chi-square Test)

- Imagine a world where variable X and variable Y are perfectly independent.

- **The Rule of Independence**: If two events are independent, the probability of them happening together is just the product of their individual probabilities:

$$P(X \text{ and } Y) = P(X) \times P(Y)$$

- Test for homogeneity of binomial proportions

$$P(Y = 1|X = 1) = P(Y = 1|X = 2)$$

Calculating the "Expected" People E_{ij}

Number of observations falling into a specific cell (let's say Cell 11) in this independent world:

1. **Prob(Row 1)** = (Row 1 Total) / (Grand Total n)
2. **Prob(Column 1)** = (Column 1 Total) / (Grand Total n)
3. **Expected Count (E_{11})** = Grand Total n × Prob(Row 1) × Prob(Column 1)

$$E_{ij} = n \times \frac{n_{i+}}{n} \times \frac{n_{+j}}{n} = \frac{n_{i+} \times n_{+j}}{n}$$

Pearson 卡方检验 (Chi-square Test)

- Hypothesis

$$H_0 : \pi_{ij} = \pi_{i+}\pi_{+j} \quad \text{for } i = 1, 2 \text{ and } j = 1, 2$$

$$H_1 : \pi_{ij} \neq \pi_{i+}\pi_{+j} \quad \text{for some } i \text{ and } j$$

- *Pearson χ^2 test*

$$X^2 = \sum_{i=1}^{2}\sum_{j=1}^{2} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \overset{H_0, approx}{\sim} \chi^2_1$$

$$O_{ij} = n_{ij}, \quad E_{ij} = n\hat{\pi}_{i+}\hat{\pi}_{+j} = \frac{n_{i+}n_{+j}}{n}$$

- **Observed O_{ij}**: What we actually saw in our data.
- **Expected E_{ij}**: What we should have seen if X and Y were perfectly independent (The "Ideal World").
- **The difference $O_{ij} - E_{ij}$** is the "error" or "surprise" for each cell.

Chi-square is simply a measure of divergence. It measures the total deviation between the actual data we observed and a hypothetical world where the two variables have absolutely nothing to do with each other.

Pearson 卡方检验+ Yates-correction

- When all of the expected numbers (E_{11}, E_{21}, E_{12} *and* E_{22}) are greater than or equal to 5

- *Pearson* χ^2 *test*

$$X^2 = \frac{(|O_{11} - E_{11}| - 0.5)^2}{E_{11}} + \frac{(|O_{21} - E_{21}| - 0.5)^2}{E_{21}} + \frac{(|O_{12} - E_{12}| - 0.5)^2}{E_{12}} + \frac{(|O_{22} - E_{22}| - 0.5)^2}{E_{22}} \overset{d}{\sim} \chi^2_1.$$

The ``-0.5" is a continuity correction (Yates correction).

- **Observed O_{ij}**: What we actually saw in our data.
- **Expected E_{ij}**: What we should have seen if X and Y were perfectly independent (The "Ideal World").
- **The difference $O_{ij} - E_{ij}$** is the "error" or "surprise" for each cell.

Critical value: χ^2_$\{1, 1-\alpha\}$

p-value = Pr($\chi_1^2 \geq X^2$)

Example continued

- 2x2 contingency table (observed)

	Age at First Birth		
Breast Cancer Status	$\geq$30	<30	Total
With	683	2537	3220
Without	1498	8747	10245
Total	2181	11284	13465

- Whether the underlying probability of having an age at first birth of $\geq$30 is different in the two groups
  - $H_0 : p_1 = p_2$ versus $H_0 : p_1 \neq p_2$
  - p_1 and p_2 are the probabilities that age at first birth is $\geq$30 in women with and without breast cancer, respectively
  - H_0 : Age at birth is independent of breast cancer status

- 2x2 contingency table (expected)

Breast Cancer Status	Age at First Birth ≥30	Age at First Birth <30	Total
With	521.6	2698.4	3220
Without	1659.4	8585.6	10245
Total	2181	11284	13465

- Use chi-square test because all expected numbers are ≥ 5
- Test statistic under H_0

$$X^2 = \frac{(|683-521.6|-0.5)^2}{521.6} + \frac{(|1498-1659.4|-0.5)^2}{1659.4} + \frac{(|2537-2698.4|-0.5)^2}{2698.4} + \frac{(|8747-8585.6|-0.5)^2}{8585.6} = 77.89 \overset{d}{\sim} \chi^2_1$$

- At $\alpha = 0.05$, the critical value is $\chi^2_{1,0.95} = 3.84$ and the p-value is $\Pr(\chi^2_1 \geq 77.89) = 1.0920957 \times 10^{-18}$

Fisher精确检验

- Chi-squared test is based on approximations, which is suitable only when all the cell frequencies are decent.
- Need exact test with existence of small cell frequencies
- For example, rare disease

	Variable X		
Variable Y		Yes	No	
	Yes	a	b	a+b
	No	c	d	c+d
		a+c	b+d	N

$$P(\text{Observed Table}) = \frac{\binom{a+c}{a}\binom{b+d}{b}}{\binom{N}{a+b}}$$

- Fisher's Exact Test assumes both row totals and column totals are fixed.
- Under the Null Hypothesis (no association), the specific arrangement of successes across rows is purely random, subject to these marginal constraints.
- **Intuition**: The Urn Model (Sampling Without Replacement)

Fisher精确检验

- Fisher's exact test: (one-sided) **p-value** = The total probability of the observed table plus all other tables that are "more extreme".

$$P\text{-value} = \sum P(\text{Table}_i) \quad \text{for all tables where } P(\text{Table}_i) \leq P_{obs}$$

- Takes discrete values for small samples

Fisher精确检验

	Variable X		
Variable Y	Yes	No	
Yes	7	2	9
No	2	8	10
	9	10	19

- What are the possible outcomes?
- What are the more extreme cases?

```r
# 1. 创建 2x2 列联表数据
# 使用 matrix 函数，byrow = TRUE 表示按行输入数据
lung_cancer_data <- matrix(c(688, 650,
                             21, 59),
                           nrow = 2,
                           byrow = TRUE)

# 2. 为行和列命名，让输出结果更易读
rownames(lung_cancer_data) <- c("Smoking_Yes", "Smoking_No")
colnames(lung_cancer_data) <- c("Cases", "Controls")

# 打印表格查看是否正确
print("--- Observed Table (Oij) ---")
print(lung_cancer_data)

# 3. 进行 Pearson 卡方检验 (Chi-square Test)
chi_test_result <- chisq.test(lung_cancer_data, correct = FALSE)

print("--- Pearson Chi-square Test Results ---")
print(chi_test_result)

print("--- Expected Table (Eij) ---")
print(chi_test_result$expected)

# 4. 进行 Fisher 精确检验 (Fisher's Exact Test)
fisher_test_result <- fisher.test(lung_cancer_data)

print("--- Fisher's Exact Test Results ---")
print(fisher_test_result)
```

```
> print("--- Observed Table (Oij) ---")
[1] "--- Observed Table (Oij) ---"
> print(lung_cancer_data)
            Cases Controls
Smoking_Yes   688      650
Smoking_No     21       59
```

```
> print(chi_test_result)

        Pearson's Chi-squared test

data:  lung_cancer_data
X-squared = 19.129, df = 1, p-value = 1.222e-05
```

```
>
> print("--- Expected Table (Eij) ---")
[1] "--- Expected Table (Eij) ---"
> print(chi_test_result$expected)
            Cases Controls
Smoking_Yes   669      669
Smoking_No     40       40
```

```
> print("--- Fisher's Exact Test Results ---")
[1] "--- Fisher's Exact Test Results ---"
> print(fisher_test_result)

        Fisher's Exact Test for Count Data

data:  lung_cancer_data
p-value = 1.476e-05
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 1.755611 5.210711
sample estimates:
odds ratio
  2.971634
```

Breast Cancer Example

A hypothesis has been proposed that breast cancer in women is caused in part by events that occur between the age at menarche (the age when menstruation begins) and the age at first child birth.

	Age at First Birth					
	< 20	20–24	25–29	30–34	≥ 35	**Total**
Case	320	1206	1011	463	220	3220
Control	1422	4432	2893	1092	406	10245
Total	1742	5638	3904	1555	626	13465
% Cases	0.184	0.214	0.259	0.298	0.351	0.239

I × J 列联表（Contingency Table ）

- Underlying probability distribution of X (smoking) and Y (cancer)

  From yes/no to (0, 1-3 per week, 3-7/week, >7 week)

- Joint distribution: π_{ij}, probability of cell (i, j)
- Marginal distribution:

$$\pi_{i+} = \sum_{j=1}^{J} \pi_{ij}, \text{ probability of row } i$$
$$\pi_{+j} = \sum_{i=1}^{I} \pi_{ij}, \text{ probability of column } j$$

- Conditional distribution:

$$\pi_{j|i} = \pi_{ij}/\pi_{i+}, \text{ distribution of } j \text{ given } i$$

- Aim: compare conditional distribution of Y at various levels of X

	Column			
Row	1	$\cdots$	J	Total
1	$\pi_{11}(\pi_{1\|1})$	$\cdots$	$\pi_{1J}(\pi_{J\|1})$	π_{1+}
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
I	$\pi_{I1}(\pi_{1\|I})$	$\cdots$	$\pi_{IJ}(\pi_{J\|I})$	π_{I+}
Total	π_{+1}	$\cdots$	π_{+J}	π_{++}

Independence: $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i and j

Homogeneity of conditional distributions:

$\pi_{j|i=}\pi_{+j}$ for all i and j

Samplings for contingency tables

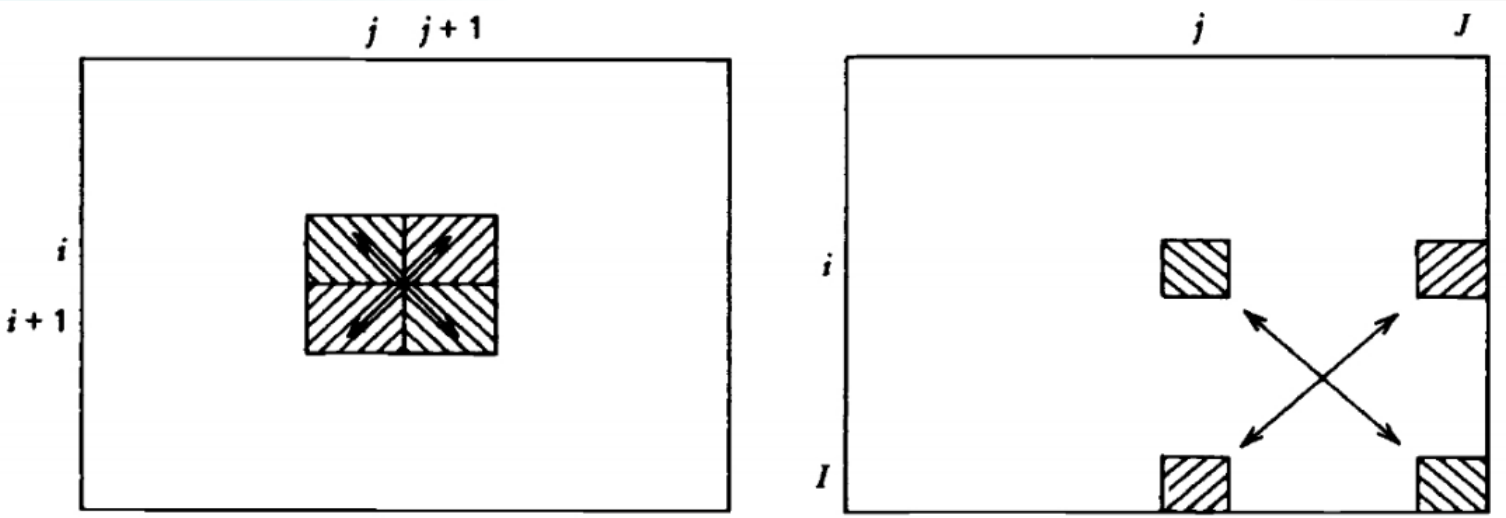
- **Independent multinomial sampling**: One margin is fixed
  - **Prospective studies** condition on row totals n_{i+}; Cohort Studies and Clinical Trials
  - **retrospective studies** condition on column totals n_{+j}
- Both row and column margins are fixed: **hypergeometric sampling**

Odds ratio in $I \times J$ tables

1) local odds ratios: $\theta_{ij} = \frac{\pi_{ij}\pi_{i+1,j+1}}{\pi_{i,j+1}\pi_{i+1,j}}$, 2)
$\theta_{ij} = \frac{\pi_{ij}\pi_{IJ}}{\pi_{Ij}\pi_{iJ}}$, $i = 1, ..., I-1$, $j = 1, ..., J-1$

Either of them can describe any associations in an $I \times J$ table (given marginal distributions)

Independence is equivalent to all these odd ratios equal 1



独立性检验

- Independence in this context is often called homogeneity of the conditional distributions
- *X* and *Y* are independent, $\Leftrightarrow \pi_{j|1} = \cdots = \pi_{j|I}$, for all j
- Can interpret the independence in terms of product of marginal probabilities

$$\pi_{ij} = \pi_{i+}\pi_{+j} \text{ for all } i \text{ and } j$$
$$\Leftrightarrow \pi_{j|1} = \cdots = \pi_{j|I} \text{ for all } j$$

- Test the homogeneity of conditional distributions

	Column			
Row	1	$\cdots$	J	Total
1	$\pi_{11}(\pi_{1\|1})$	$\cdots$	$\pi_{1J}(\pi_{J\|1})$	π_{1+}
$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$
I	$\pi_{I1}(\pi_{1\|I})$	$\cdots$	$\pi_{IJ}(\pi_{J\|I})$	π_{I+}
Total	π_{+1}	$\cdots$	π_{+J}	π_{++}

Consider the new notation $\pi_j(x) = P(Y = j|X = x)$

Although the interpretation is different, use the same Pearson X^2 test and the LR test

Pearson 卡方检验 (Chi-square Test)

- Hypothesis

H_0 : reduced model $\pi_{ij} = \pi_{i+}\pi_{+j}$, for all i and j

H_a : full model $\pi_{ij} \neq \pi_{i+}\pi_{+j}$, for some i and j

- *Pearson χ^2 test*

$$X^2 = \sum_{i=1}^{I}\sum_{j=1}^{J} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \overset{H_0, approx}{\sim} \chi^2_{(I-1)\times(J-1)}$$

$$O_{ij} = n_{ij}, E_{ij} = n\hat{\pi}_{i+}\hat{\pi}_{+j} = n_{i+}n_{+j}/n$$

Critical value: χ^2_{$(I-1)\times(J-1), 1-\alpha$}

Test is valid when: (rule of thumb)

No more than 1/5 of cells have expected values < 5

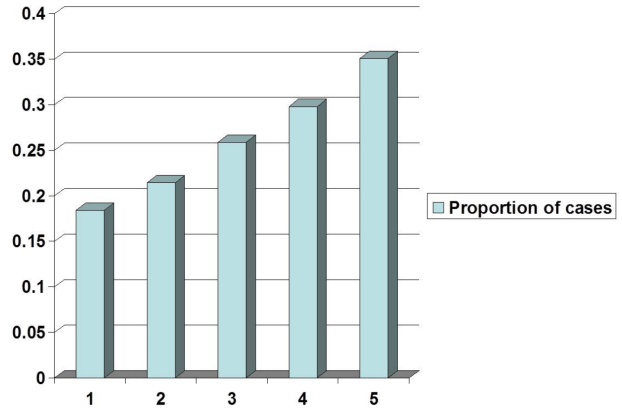
No cells have epxected values < 1

Extension (for external reading)

An Introduction to Categorical Data Analysis 2nd Edition, by Alan Agresti (2007)

Chi-squared test for trend in binomial proportions

	Age at First Birth					
	< 20	20–24	25–29	30–34	≥ 35	Total
No. of Cases (x_i)	320	1206	1011	463	220	3220
Total (n_i)	1742	5638	3904	1555	626	13465
p_i	0.184	0.214	0.259	0.298	0.351	0.239



Paired Data (more generally matched data•

One random variable measured on the same unit under different conditions
One random variable measured on the same unit in different locations
One random variable measured by two different readers on the same unit
Matched samples)

Why do we need Non-parametric Tests?

- *What we just did* **(Contingency Tables):** Counted data in categorical bins (e.g., Yes/No). We didn't assume the data followed a bell curve.
- *What you learned before* **(Parametric Tests):** Compared continuous numbers (e.g., Blood Pressure).
- *Crucial Assumption***:** The data follows a strict, well-behaved mathematical distribution.

Are we testing the same Null Hypothesis H_0?

Chi-square *:* Their category proportions are the same (Independence).

T-test *:* Their population *means* are the same.

Non-parametric *:* Their underlying *overall distributions* are the same.

How? By transforming the raw, messy numbers into robust, distribution-free metrics—like **Ranks.**

非参数检验回顾

The Limitations of Parametric Models

Traditional parametric tests (like the Student's t-test or ANOVA) and standard generalized linear models are heavily reliant on strict distributional premises—most commonly, the assumption of normality or binomial distributions. When these assumptions fail, p-values become severely biased.

Common Pain Points in Biostatistics:

Small Sample Sizes

In early-stage clinical trials or rare disease studies, small 'N' prevents the reliable application of the Central Limit Theorem, making parametric tests invalid.

Extreme Skewness

Biomarker expressions (e.g., cytokine levels) often exhibit extreme outliers that cannot be normalized via log-transformations, heavily distorting mean-based tests.

Ordinal Outcomes

Clinical data is frequently recorded on ordinal scales (e.g., Mild, Moderate, Severe). Calculating a formal "mean" for such categorical rankings is mathematically inappropriate.

The Wilcoxon Rank-Sum Test

Designed to compare the median differences between two independent samples. It serves as the definitive non-parametric alternative to the independent two-sample t-test.

Null Hypothesis H_0: The two independent samples come from populations with identical distributions (or equal medians).

The Mathematical Mechanism

- **Pooling & Sorting:** Data from both groups are pooled together and arranged in ascending order, from smallest to largest.
- **Ranking:** Each data point is assigned a numerical "Rank" (1 for the smallest, N for the largest).
- **Summation:** We calculate the sum of the ranks for each group independently.
- **Evaluation:** If one group's values are systematically larger, its cumulative rank-sum will be significantly higher than expected by random chance.

The Wilcoxon Rank-Sum Test

Scenario: Comparing a new Targeted Therapy vs. Traditional Chemotherapy by measuring the percentage change in tumor volume (RECIST criteria). Because of one severe outlier, the mean heavily penalizes the Targeted Therapy, making the t-test fail. The Wilcoxon test remains highly robust.

Patient ID	Treatment Group	Tumor Volume Change	Statistical Rank
P01	Targeted Therapy	**- 90%** (Excellent Response)	**1** (Smallest value)
P02	Chemotherapy	- 45%	2
P03	Targeted Therapy	- 30%	3
P04	Chemotherapy	- 10%	4
P05	Targeted Therapy	**+ 150%** (Extreme Outlier)	**5** (Caps the outlier's influence)

The Non-Parametric Advantage: The +150% growth completely distorts the mean of the Targeted group. By converting values to ranks, +150% simply becomes "Rank 5", preventing one abnormal biological response from invalidating the entire study.

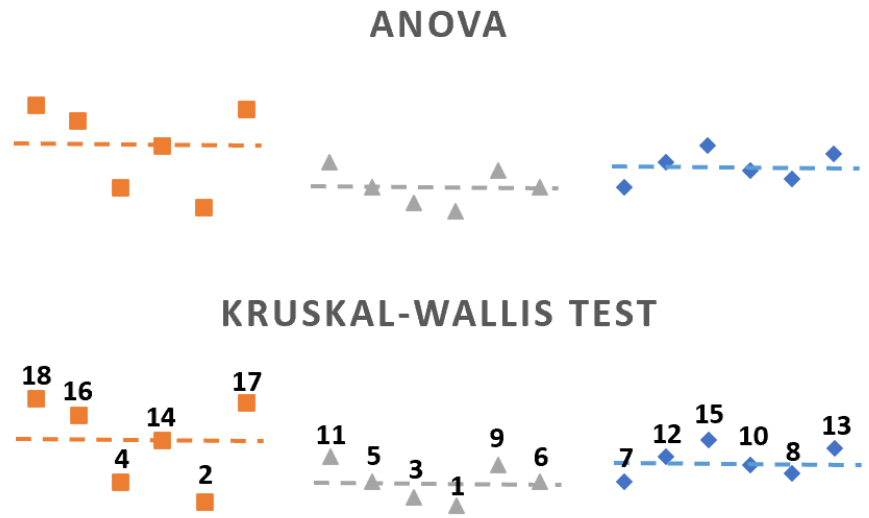
The Kruskal-Wallis H Test

- The Kruskal-Wallis test is the non-parametric extension of the Wilcoxon test for comparing **three or more independent samples**. It replaces the One-Way ANOVA.

- Instead of comparing means, it ranks the entire pooled dataset and evaluates how much the average rank of each individual group deviates from the expected global average rank.

- **Statistic Construction** (***H***): Similar to Wilcoxon, it ranks all pooled data. The H statistic calculates the variance of the ranks—specifically, how much the average rank of each individual group deviates from the overall global average rank of the entire dataset.

- For adequately large samples (typically $n_i \geq 5$), the H statistic closely follows a Chi-Square (χ^2) distribution with $k - 1$ degrees of freedom.



Wilcoxon Rank-Sum (U)

For comparing **2** independent groups

$$U = R_1 - \frac{n_1(n_1+1)}{2}$$

- R_1 = Sum of ranks for Group 1
- n_1 = Sample size of Group 1
- The subtracted term $\frac{n_1(n_1+1)}{2}$ represents the absolute minimum possible rank sum for a group of this size.

Kruskal-Wallis (H)

For comparing $k \geq 3$ independent groups

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3\left(N+1\right)$$

- N = Total sample size across all groups
- k = Total number of groups
- R_i = Sum of ranks for the i-th group
- n_i = Sample size of the i-th group

By converting continuous, skewed biological data into discrete ranks, we neutralize outliers. These test statistics mathematically evaluate the deviation of rank sums from their expected null values.

The Kruskal-Wallis H Test

Scenario: Measuring IL-6 Cytokine levels (pg/mL) across 3 breast cancer subtypes. We rank all 9 patients globally, then sum the ranks (R_i) per group to calculate H.

Patient	Subtype	IL-6 Level	Global Rank
P1, P2, P3	**Luminal A**	10, 12, 15	1, 2, 3 ($R_1 = 6$)
P4, P5, P6	**Luminal B**	25, 30, 45	4, 5, 7 ($R_2 = 16$)
P7, P8, P9	**TNBC**	40, 80, 120	6, 8, 9 ($R_3 = 23$)

Calculating the H Statistic

Given: $N = 9, n_i = 3$

$$H = \frac{12}{9(10)}\left(\frac{6^2}{3} + \frac{16^2}{3} + \frac{23^2}{3}\right) - 3(10)$$

$$H = \frac{12}{90}\left(12 + 85.3 + 176.3\right) - 30 \approx 6.48$$

Conclusion: With df = 2, the critical χ^2 value at $\alpha = 0.05$ is 5.99. Since $6.48 > 5.99$, we **reject the null hypothesis**. TNBC has significantly higher IL-6 levels.

现代补充：医学AI中的混淆矩阵

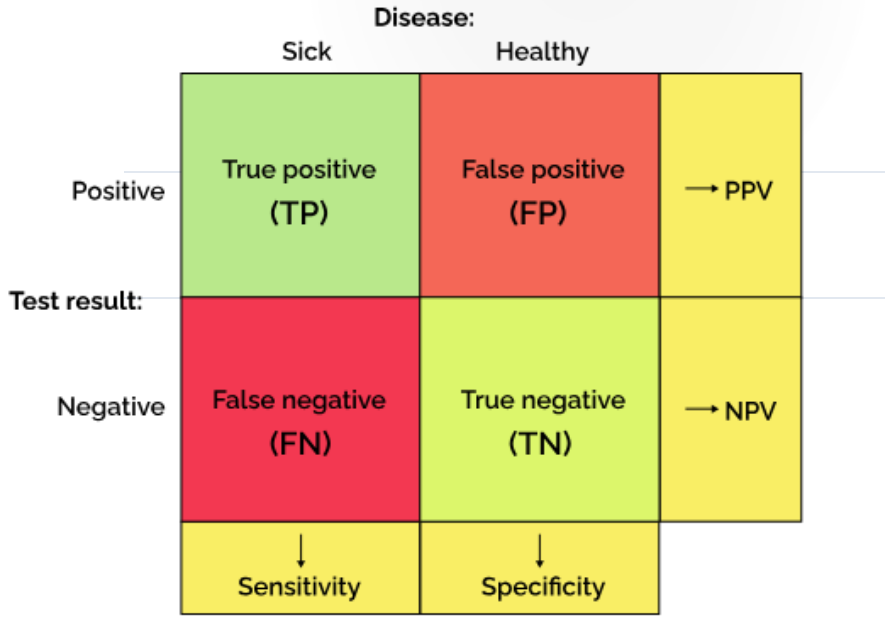
In diagnostics, the 2x2 table maps directly to classification performance:

Prediction / Actual	Cases (Positive)	Controls (Negative)
Positive	True Positive (TP)	False Positive (FP)
Negative	False Negative (FN)	True Negative (TN)

Sensitivity (Sensitivity): $P(\text{Test} + | \text{Disease} +)$

Specificity (Specificity): $P(\text{Test} - | \text{Disease} -)$

Accuracy= (TP+TN)/N



Why accuracy can fail in medical AI

Rare disease example: prevalence = 1%

Model A: predict everyone healthy

	Disease +	Disease -
Test +	0	0
Test -	10	990

$$\text{Accuracy} = \frac{990}{1000} = 99\%$$

$$\text{Sensitivity} = \frac{0}{10} = 0$$

$$\text{Specificity} = \frac{990}{990} = 1$$

Model B: detects most cases

	Disease +	Disease -
Test +	8	50
Test -	2	940

$$\text{Accuracy} = \frac{8 + 940}{1000} = 94.8\%$$

$$\text{Sensitivity} = \frac{8}{10} = 80\%$$

$$\text{Specificity} = \frac{940}{990} = 94.9\%$$

A higher accuracy does not necessarily mean a better diagnostic model.
In rare-disease screening, sensitivity is often prioritized because false negatives are costly.